

FIONA BECHTOLD

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EDUCATION

University of Victoria

2021-August 2026

Bachelor of Engineering: Biomedical Engineering, Electrical Stream

Jamie Chien-Ming Wu Award for Achievement in Biomedical Engineering | Uvic Excellence Scholarship | Faculty of Computer Science and Engineering Dean's Entrance Scholarship | Women in Engineering and Computer Science Entrance Scholarship

WORK EXPERIENCE

Victoria Harbour Ferries

Dock Staff/Sales Associate

April-September 2024, April 2026-present

- Maintained strong product knowledge to provide clear, confident recommendations to customers
- Responded to customer questions and resolved issues in real time, demonstrating strong problem-solving and communication skills

Salvica Health (Previously Salyx Medical)

Project Manager

May-December 2025

- Coordinated cross-functional engineering activities, ensuring timely delivery of components and alignment with technical and user requirements
- Communicated technical information clearly to both technical and non-technical stakeholders to support product adoption and usability
- Supported rapid issue resolution by identifying and escalating blockers in a fast-paced development environment

Island Health

Island Health Infection Prevention and Control (IPAC) Co-op Student

January-August 2023

- Provided real-time coaching and education to healthcare staff and physicians on proper procedural practices in clinical environments
- Collaborated with multidisciplinary hospital teams to improve clinical workflow adoption and user compliance
- Provided direct, on-the-spot feedback during clinical practice observations, supporting performance improvement under operational pressure
- Worked within 7 hospital environments across Vancouver Island, adapting communication style to different clinical stakeholders and operational contexts and formed professional relationships with a diverse set of healthcare practitioners
- Trained 50+ healthcare professionals in IPAC specific software

PROJECTS

Redesign of a Goniometer for Improved Wrist Pronation and Supination Measurements

January 2026 – May 2026

- Conducted a clinical workflow analysis in a hospital environment to evaluate usability and limitations of an existing wrist goniometer
- Collaborated with clinicians to trial and assess device performance in real patient-care settings
- Provided hands-on training and onboarding to clinical staff for safe and effective device use
- Supported iterative redesign efforts to improve accuracy, usability, and clinical workflow integration

fNIRS Neurofeedback System

January 2026-May 2026

- Designed and implemented an integrated biomedical system (hardware + software) for functional near-infrared spectroscopy (fNIRS)-based neurofeedback
- Validated system performance through iterative testing in a biomedical device development context
- Collaborated with healthcare professionals for feedback to encourage iterative design for the clinical environment

TECHNICAL SKILLS AND COMPETENCIES

Certifications: Class 5 Drivers License (clean driving record); Canadian and American dual citizenship; ROC-M marine radio license

Embedded and Electrical: PCB Design (KiCad and Altium Designer); firmware development (C, C++, MicroPython); oscilloscope operation; hardware troubleshooting and fault isolation; circuit simulation (LTspice); soldering and hardware prototyping

Programming and Software: Microsoft Office Suite; Python, MATLAB; RStudio, computer vision; Git version control

Mechanical: SOLIDWORKS; Siemens NX; 3D Printing; GD&T; engineering drawings